Project Documentation

Product Inventory Management

1. Introduction

- Project Title: Product Inventory Management
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2. Project Overview

- Type: Backend REST API Project
- Technologies: Node.js, Express.js, MongoDB, JWT Authentication
- The Product Inventory Management System is a backend web application designed to manage product data through REST APIs.
- The system allows users to register, log in securely, and manage their own product inventory.
- Each user can create, view, update, and delete products.
- Authentication is implemented using JSON Web Tokens (JWT) to ensure that only authorized users can access protected routes.

3. Architecture

- Backend: Node.js with Express.js
- Database: MongoDB using Mongoose ODM
- Authentication: JWT (JSON Web Token)
- Architecture Pattern: MVC (Model – Controller – Route)

4. Setup Instructions

Prerequisites:

- Node.js
- Npm

- MongoDB
- VisualStudioCode

5. Installation Steps:

- Open terminal in the project folder
- Run: npm install
- Configure environment variables in .env file
- Start server using: npm start

6. Folder Structure:

```
server/
│
├── node_modules/
├── src/
│   ├── controllers/
│   │   ├── auth.controller.js
│   │   └── product.controller.js
│   ├── db/
│   │   └── db.js
│   ├── middlewares/
│   │   └── auth.middleware.js
│   ├── models/
│   │   ├── User.js
│   │   └── Product.js
│   ├── routes/
│   │   ├── auth.routes.js
│   │   └── product.routes.js
│   └── server.js
│
├── .env
├── package.json
└── package-lock.json
```

6. API Documentation

- Authentication APIs:
  - POST /api/v1/auth/register – Register a new user
  - POST /api/v1/auth/login – Login and receive JWT token
- Product APIs:
  - POST /api/v1/products – Create new product
  - GET /api/v1/products – Get all products for logged user
  - PUT /api/v1/products/:id – Update product
  - DELETE /api/v1/products/:id – Delete product

7. Authentication

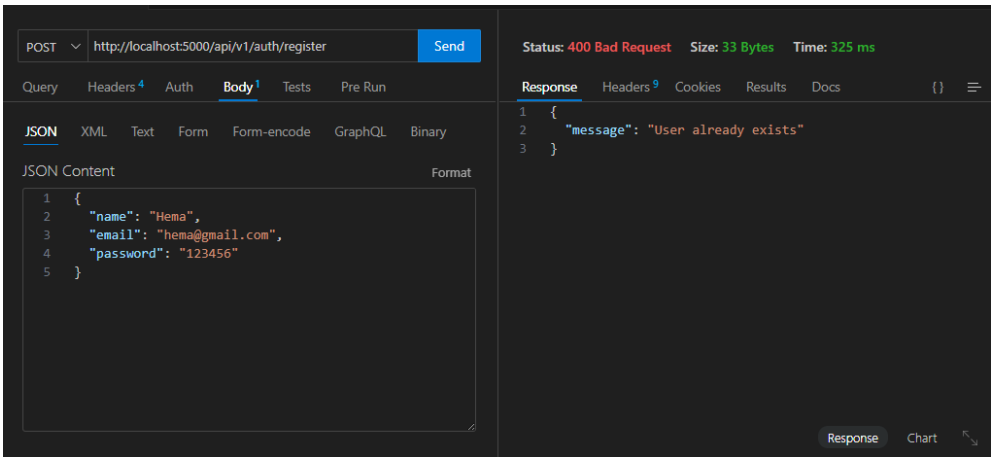
- JWT authentication is used to secure protected routes.
- After a successful login, a token is generated and must be sent in the Authorization header as 'Bearer <token>' to access product APIs.

8. Testing

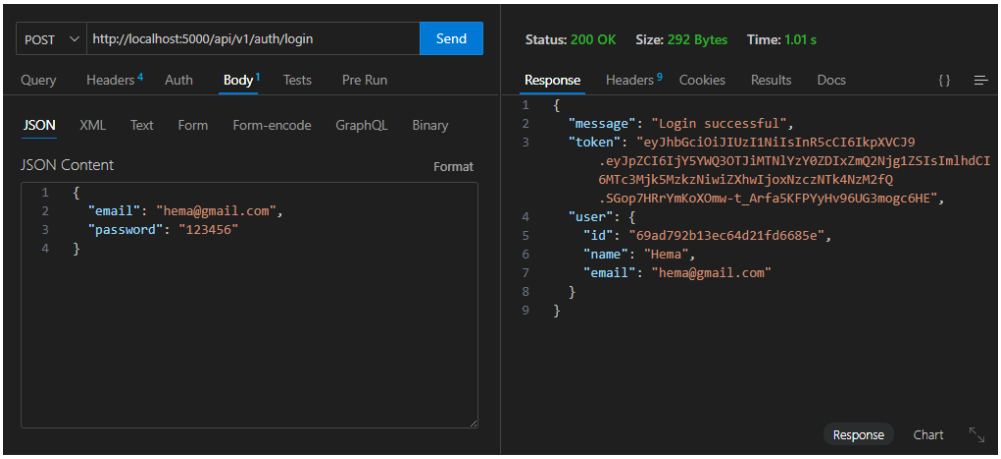
- APIs were tested using Thunder Client extension in Visual Studio Code.
- Testing included user registration, login authentication, product creation, product retrieval, update, and deletion.

9. Screenshots

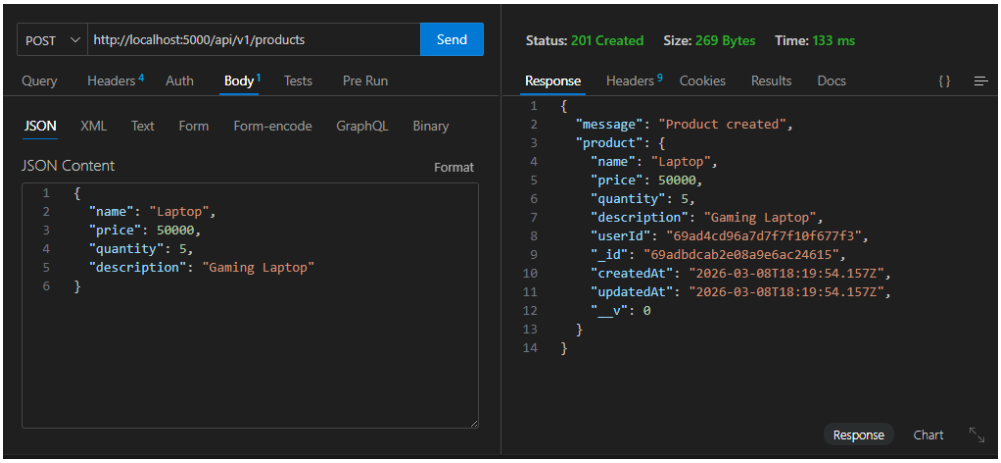
- Register API response



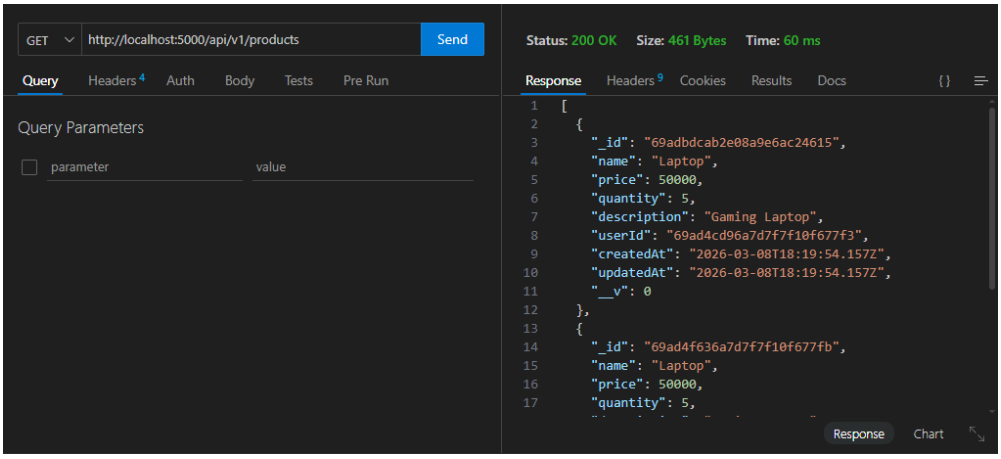
- Login API response



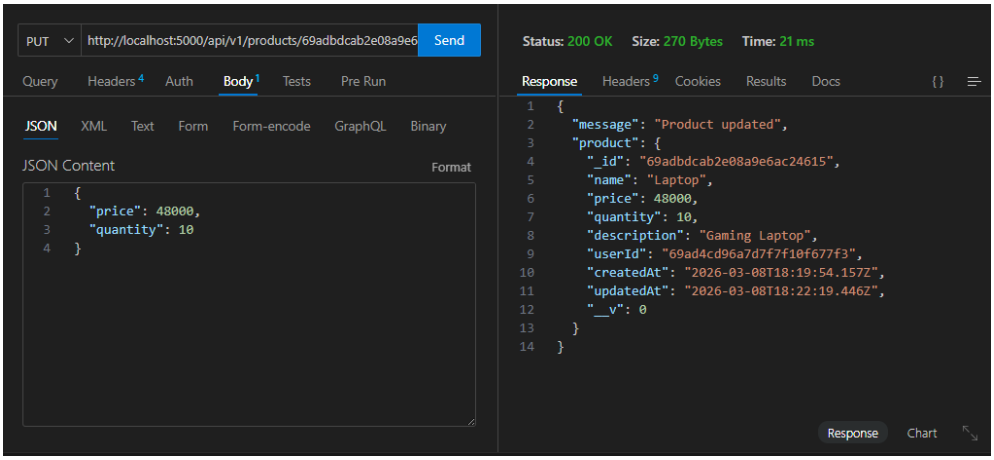
- Create Product API



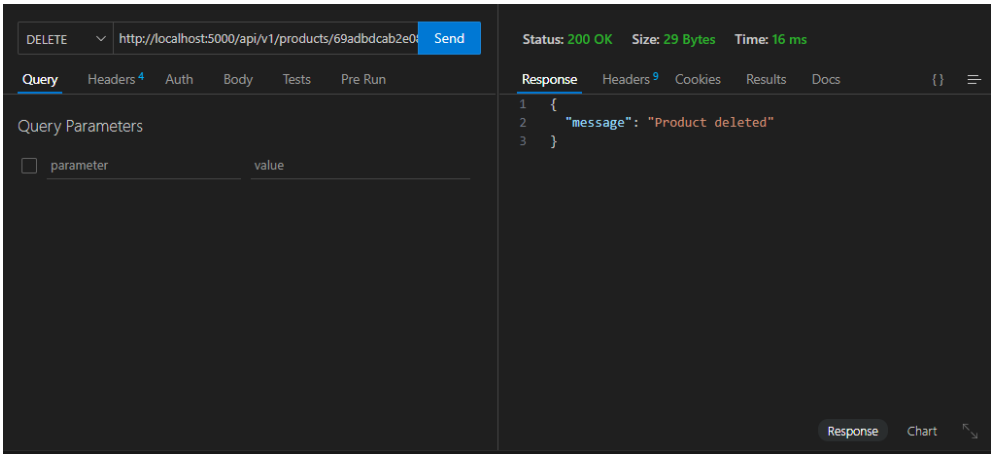
- Get Products API



- Update Product API



- Delete Product API



10. Known Issues

No major issues reported during development and testing.

11. Future Enhancements

- Add frontend interface using React
- Implement product image upload
- Add pagination and advanced search
- Add role-based access control